

NICHOLAS SAROIU

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EDUCATION

- 2022-2026 **Honours Bachelor of Science**, Computer Science Major, Mathematics Minor
University of Toronto, Faculty of Arts and Science, Toronto, Canada
Relevant Coursework: Software Design; Data Structures and Analysis; Algorithm Design, Analysis & Complexity; Introduction to Databases; Introduction to Machine Learning; Neural Networks and Deep Learning; Computation Linguistics; Operating Systems; Computer Architecture.

WORK/VOLUNTEER EXPERIENCE

- SEPT 2025 **Student Researcher**
— PRESENT *Lee Language Lab*, Toronto, Ontario
Conducting research on applications of Punctuation Restoration as an unsupervised learning objective for language models. Investigating language-agnostic performance gains, with a focus on improving model effectiveness for low-resource languages. Collaborating with a team of three students on experimental design and result analysis.
- MAY 2025 **Jr. Software Engineer**
— AUG 2025 *Dibbly Inc.*, Oakville, Ontario
Implemented comprehensive test coverage for existing codebase using Jest and Pytest automated testing frameworks. Contributed to development of full-stack AI applications within an agile team of ten developers using React.js, Next.js, and Python.
- JUNE 2024 **Frontend Developer**
— OCT 2024 Toronto, Ontario
Built interactive real-time dashboard for residential battery systems displaying performance metrics and energy analytics. Developed responsive React.js components with Chart.js data visualization to help homeowners optimize energy consumption. Collaborated with independent development team to deliver production-ready application with mobile-responsive design.

PROJECTS

- OCT 2025 **Understanding Transformers via Causal Tracing**
Reproduced and analyzed interpretability experiments from the NeurIPS 2022 ROME paper "*Locating and Editing Factual Associations in GPT*". Applied causal tracing interventions on attention heads and intermediate activations to study how transformer models store and recall factual knowledge, using targeted ablation and restoration to identify influential internal components.
- NOV 2024 **Models to Predict Student Accuracy**
Built three machine learning models (K-Nearest Neighbors, Item-Response Theory, and Autoencoder Neural Network) to predict student performance. Implemented data preprocessing pipeline and ensemble methods to achieve 70% validation accuracy. Utilized PyTorch for neural networks and scikit-learn for traditional ML algorithms with cross-validation optimization.
- NOV 2023 **TTC Tracker**
<https://github.com/nsaroiu/TTC-Tracker>
Built real-time web application tracking Toronto TTC vehicles across 20+ routes with arrival predictions. Developed React.js frontend with interactive maps and Java Spring Boot backend integrating TTC's GTFS data feed.

PROGRAMMING SKILLS

Python, Java, C, Javascript, Typescript, SQL, React.js, Next.js, Spring Boot, Django, PyTorch, NumPy, HTML/CSS, Docker, Git, LaTeX, MIPS Assembly